Jans Solano

Education

Pontifical Catholic University of Peru (PUCP) *Mar 2019 – Dec 2024*
BSc and Professional Title in Mechatronics Engineering

- Grade: 17.74/20 – **Rank: 1st** out of 80 students
- Thesis: Wheeled-Legged Navigation with **RL Locomotion and Fall Recovery** *(advised by Prof. Diego Quiroz).*

University of Stuttgart *Oct 2023 – Mar 2024*
Exchange Student in Computer Science

- Grade: 1.3 (scale 1–6, 1 best) – Coursework: ML, Foundation Models, Computer Vision, DL for NLP.

Research Experience

Yale University *Remote, USA*
Research Assistant (advised by Prof. Josue Ortega) *Jan 2025 – May 2025*

- Fine-tuned **VLMs** (LLaMA, Qwen) on fMRI for clinical-variable prediction.
- Built an LLM predictor on a pretrained **fMRI encoder** achieving **18% lower MSE** than the VLM baseline.
- Aligned fMRI and text in a **shared vocabulary** via an **adversarial VQ autoencoder** trained on 6,700 h of fMRI, retaining **89%** of baseline R^2.

Max Planck Institute for Intelligent Systems (MPI-IS) *Stuttgart, Germany*
Research Intern (advised by Prof. Katherine Kuchenbecker) *Jul 2024 – Sep 2024*

- Adapted Minsight, a vision based **soft tactile sensor** by embedding a microphone, raising fabric recognition from **41%** to **95%**.
- Developed **motion patterns** for a dexterous hand and a 3-axis machine for texture data collection.
- Trained **temporal models**; proposed fusing internal and external audio, which gave the top accuracy (**97.8%**).

Pontifical Catholic University of Peru (PUCP) *Lima, Peru*
Research Assistant (advised by Prof. Diego Arce) *Aug 2022 – Aug 2023*

- **Benchmarked 4 classic and DRL algorithms** for autonomous navigation of mobile robots in simulation.
- Built the ROS navigation stack; evaluated **social navigation** across **18 pedestrian-behavior scenarios**.
- Trained and compared **mapless DRL agents** with discrete and continuous actions.

Industry Experience

Robot.com (formerly Kiwibot) *Hybrid, China-USA*
Robotics Engineer *Aug 2025 – Present*

- Developed **Isaac Sim digital twins** for robotic platforms, including tabletop systems and wheeled humanoids.
- Built humanoid **data-collection** and VR/puppet-style **teleoperation** pipelines for a fleet of **20+ robots**.
- Fine-tuned in-house **VLA models** and **Physical Intelligence's $\pi_{0.7}$** (company partnership), improving both with **human-in-the-loop** corrections.

Bosch Center for Artificial Intelligence (BCAI) *Renningen, Germany*
Working Student – Educational AI Robotics Demonstrators *Oct 2023 – Mar 2024*

- Optimized an **IL RC car** with better teleop/inference, resulting in a policy that cut lap time by **60%**.
- Trained an **LSTM autoencoder** as a proof of concept to detect faulty dice from their impact sound.
- Built RC car's **collision-sensing bumper** and a servo/conveyor rig that **automated manual** audio recording.

Publications

- **Solano J**, Quiroz D. "*Stand, Walk, Navigate: Recovery-Aware Visual Navigation on a Low-Cost Wheeled Quadruped*". **IROS 2025 - Workshop on Wheeled-Legged Robotics**. [Paper]
- Andrussow I, **Solano J**, Richardson B, Martius G, Kuchenbecker K. "*Adding internal audio sensing to internal vision enables human-like in-hand fabric recognition with soft robotic fingertips*" (**Oral**). **Humanoids 2025**. [Website], [Paper]
- **Solano J**, Cisneros J, Sarmiento L, Quispe G, Hermitaño A, Quiroz D, Balbuena J. "*Design and Implementation of an Inspection Robot for Crack Detection in Flooded Pipes.*" **IEEE INTERCON 2023**. [Paper]
- Arce D, **Solano J**, Beltrán C. "*A Comparison Study between Traditional and Deep-RL Algorithms for Indoor Autonomous Navigation in Dynamic Scenarios.*" **Sensors 2023**. [Paper]

Competitions

European Rover Challenge *Remote, Poland*
Computer Vision & Autonomous Navigation Member *2023, 2025*

- (2023) Developed algorithms for **ArUco 3D pose estimation** and robot arm motion planning.
- (2025) Implemented 3D reconstruction and SLAM modules integrating depth and LiDAR data.

Latin American Space Challenge *São Paulo, Brazil*
Electronics and Controls Lead *2021, 2023*

- (2021) Designed and programmed the electronics system for a CanSat mini-satellite.
- (2023) Developed the **Go-to-goal PID navigation algorithm** for a comeback rover CanSat.

Teaching and Mentorship

Pontifical Catholic University of Peru (PUCP) *Oct 2025 - Present*
Lecturer – ROS2 Specialization Program
Three courses (Introduction to ROS 2, Mobile Robot Navigation, and Robot Manipulation), focused on applied autonomy using ROS 2.

Mision Tech *Aug 2025*
Mentor – Robotics Bootcamp
Mentored high school students through hands-on robotics and programming challenges.

Pontifical Catholic University of Peru (PUCP) *Mar 2025 – Mar 2026*
Teaching Assistant – Robotics and AI (WS 25), Autonomous Vehicles (SS 26)
Assisted lectures and labs on kinematics, navigation, and AI, and helped test educational robotics modules.

Teens in AI *Mar 2022*
Mentor – Women's Day Hackathon
Taught participants core machine learning concepts while mentoring teams building AI prototypes solutions.

Honors and Awards

First in Class *2025*
Awarded distinction at the graduation ceremony for ranking 1st out of 80 Mechatronics graduates in the 2024 class.

CaCTüS Cohort *2024*
9 selected from 700+ applicants worldwide to participate in a fully funded internship at the Max Planck Institute. [Link]

DARI Scholarship *2023*
Awarded an exchange scholarship for academic excellence. Ranked 1st among PUCP exchange applicants. [Link]

Latin American Space Challenge *2023*
Awarded 1st prize among 40+ teams from Latin America and Asia in the CanSat and Overall Satellite categories. [Link]

European Rover Challenge *2023*
Finalists in the European Rover Challenge Remote 2023 [Link]

COAR National Scholarship *2018*
Awarded full bachelor scholarship. Ranked 1st among 400+ applicants from high performing schools (COAR). [Link]

ONAM National Mathematics Olympiad *2017*
Awarded 1st Place in the ONAM Trilce National Mathematics Olympiad. [Link]

Leadership and Community Service

PUCP Mechatronics Students Association *2023*
Academic Affairs Representative: Supported academic community by organizing talks and compiling past exams.

PUCP Robotics Club *2022*
Project Coordinator: Coordinated multidisciplinary robotics projects and organized technical workshops for competitions.

Skills

- **Programming:** Python, C/C++, MATLAB.
- **Tools & Simulation:** ROS 1/2, Gazebo, MuJoCo, Isaac Sim/Lab, Docker, Git, Google Cloud.
- **Libraries & Frameworks:** Transformers, PyTorch, Tensorflow, OpenCV, MoveIt, Nav2.
- **Languages:** English (Fluent), Spanish (Native), German (Basic).